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Software Requirement Specification [SRS] Document On

Project Name: School Management System

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1. INTRODUCTION

1.1 Project Overview

The School Management System (SMS) is a software application designed to automate and simplify the daily administrative and academic activities of a school. It provides a centralized platform where administrators, teachers, and students can access information according to their roles. The system is intended to reduce paperwork, minimize human errors, improve record management, and make important information easier to retrieve.

1.2 Background and Problem Statement

Many schools manage student and teacher information, attendance, examinations, fees, and notices using paper files or spreadsheets. Such manual processes are time-consuming, difficult to search, and prone to data-entry and record-management errors. The proposed system addresses these problems by providing centralized digital management and role-based access.

1.3 Process Model

The School Management System will follow the **Agile Scrum Model** for software development. Agile Scrum is suitable for this project because the system contains multiple independent modules that can be developed, tested, reviewed, and improved incrementally. The project includes features such as user authentication, student management, teacher management, class and subject management, attendance, examination and results, fee management, notices, reports, and password management. These features can be divided into smaller development tasks and completed through short iterations called **Sprints**.

In each Sprint, the team will select a group of requirements from the product backlog and develop them as a working part of the system. After implementation, the completed features will be tested and reviewed. Feedback from the team or stakeholders can then be used to improve the next Sprint. This approach reduces development risk and allows problems to be identified earlier.

For this project, the development process can be organized as follows:

1. **Requirement Gathering:** Identify and prioritize the requirements of Admin, Teacher, and Student users.
2. **Product Backlog Creation:** Convert the requirements into user stories and prioritize them according to importance.
3. **Sprint Planning:** Select a small group of user stories for each Sprint.
4. **Design and Development:** Design the required interface and database components and implement the selected features.
5. **Testing:** Test each completed feature to ensure that it satisfies its acceptance criteria.
6. **Sprint Review:** Review the completed functionality and identify required improvements.

7. **Sprint Retrospective:** Discuss problems faced during the Sprint and determine how the development process can be improved.
8. **Next Sprint:** Continue developing the next set of prioritized features until the complete system is finished.
9. **Final Integration and Deployment:** Integrate all modules, perform complete system testing, fix remaining issues, and prepare the final application.

Why Agile Scrum is Suitable

Agile Scrum is appropriate for the School Management System because:

- The system contains multiple modules that can be developed independently.
- User stories can be directly converted into Sprint tasks.
- Features can be implemented and tested incrementally.
- Errors can be detected and corrected earlier.
- Changes or improvements to requirements can be handled more easily.
- The team can continuously review the progress of the project.
- Working versions of the system can be produced after each Sprint.
- It is suitable for a student group project where different team members can work on different modules.

The major development flow will therefore be:

Requirements → Product Backlog → Sprint Planning → Design & Development → Testing → Sprint Review → Improvement → Next Sprint → Final Integration

The Agile Scrum approach supports the project's existing requirement structure because the SRS already defines user stories with priorities and acceptance criteria, such as login, student management, attendance, examination results, fee management, notices, reports, and password management.

2. SCOPE & FEATURES

2.1 Role: Student

Scope 1: Account Login and Authentication

The Student can log in using valid account credentials.

The Student enters the Student ID and password on the login page. The system verifies the credentials and opens the Student Dashboard after successful authentication.

To securely access personal academic information and student functions.

Scope 2: Profile Management

The Student can view their personal and account-related profile information.

The Student opens the Profile section after login. The system retrieves the stored profile information and displays it.

To allow the Student to check the information maintained by the school.

Scope 3: Attendance Management

The Student can view their attendance information.

The Student selects Attendance from the dashboard. The system retrieves the student's attendance records from the database and displays them.

To allow the Student to monitor their attendance.

Scope 4: Examination and Result Management

The Student can view examination results and marks recorded for the student.

The Student opens the Examination/Result section. The system retrieves available examination information and results and displays them.

To allow the Student to monitor academic performance.

Scope 5: Notice Management

The Student can view notices and announcements published by the school administration.

The Student opens the Notice section. The system retrieves published notices and displays them.

To ensure the Student receives important school information.

Scope 6: Password Management

The Student can change the account password.

The Student enters the required current and new password information. The system validates the request and updates the password.

To maintain account security.

2.2 Role: Teacher

Scope 1: Account Login and Authentication

The Teacher can log in using valid teacher credentials.

The Teacher enters the Teacher ID and password. The system authenticates the credentials and opens the Teacher Dashboard.

To provide secure access to teacher-specific functions.

Scope 2: Profile and Account Management

The Teacher can access relevant account information and manage the account password.

The Teacher uses the profile and account options available from the Teacher Dashboard.

To allow secure access to teacher account information.

Scope 3: Assigned Class Management

The Teacher can view classes assigned to them.

The Teacher opens Assigned Classes and the system displays the teacher's assigned classes.

To identify teaching responsibilities.

Scope 4: Student Information Management

The Teacher can view relevant student information for assigned classes.

The Teacher selects an assigned class and accesses its student information. The system retrieves the relevant records.

To provide information needed for academic activities.

Scope 5: Attendance Management

The Teacher can record and view student attendance.

The Teacher selects an assigned class and date, records attendance, and saves it. The system stores the record.

To maintain accurate digital attendance records.

Scope 6: Examination and Marks Management

The Teacher can enter examination marks for students.

The Teacher selects the class, subject, and examination, enters marks, and submits them. The system validates and stores the marks.

To maintain examination records and support result generation.

Scope 7: Academic Report Viewing

The Teacher can view relevant academic reports available to the teacher.

The Teacher selects the relevant report and the system retrieves authorized information from stored records.

To monitor student and class academic information.

Scope 8: Password Management

The Teacher can change the account password.

The Teacher provides the current and new password information and submits the request. The system validates and updates the password.

To maintain account security.

2.3 Role: Admin

Scope 1: Account Login and Authentication

The Admin can securely log in to the administrative part of the system.

The Admin enters valid administrator credentials. The system authenticates them and opens the Admin Dashboard.

To restrict administrative functions to authorized personnel.

Scope 2: User Account Management

The Admin can manage user access and relevant user accounts.

The Admin accesses user-management functions from the Admin Dashboard and performs authorized account operations.

To control system access according to user roles.

Scope 3: Student Management

The Admin can add, view, edit, search, and delete student records.

The Admin opens Student Management, selects the required operation, enters or updates information, and saves it. The system stores the changes.

To maintain centralized and accurate student records.

Scope 4: Teacher Management

The Admin can add, view, edit, search, and delete teacher records.

The Admin opens Teacher Management, performs the required operation, and saves the information.

To maintain accurate teacher records.

Scope 5: Class Management

The Admin can manage class information.

The Admin uses Class Management to create, update, view, or remove class information.

To keep the academic structure organized.

Scope 6: Subject Management

The Admin can manage subject information.

The Admin uses Subject Management to maintain subject records.

To maintain the subjects required for academic and examination activities.

Scope 7: Fee Management

The Admin can manage student fee and fee-collection records.

The Admin records or updates fee information through Fee Management and saves it in the database.

To maintain organized and accurate fee records.

Scope 8: Notice Management

The Admin can create and publish school notices.

The Admin enters notice information and publishes it through Notice Management. Published notices become available to students.

To communicate important school information.

Scope 9: Report Management

The Admin can view or generate reports from stored system information.

The Admin selects a report and the system retrieves relevant stored data and displays it.

To monitor school activities and retrieve information efficiently.

Scope 10: Password Management

The Admin can change the administrator account password.

The Admin provides the required current and new password information. The system validates and updates the password.

To protect administrative access and maintain security.

2.4 Scope Boundary

The current scope covers functions assigned to Admin, Teacher, and Student in the provided documents. Online payment gateways, mobile applications, parent accounts, and external third-party integrations are outside the current scope because they are not specified as current system features.

3. USE CASE DIAGRAM

The use case diagram below summarizes the main interactions between the three user roles defined in the project proposal and the School Management System. Admin has the broadest management access; teachers perform academic operations such as attendance and marks; and students primarily view their own academic information and notices.

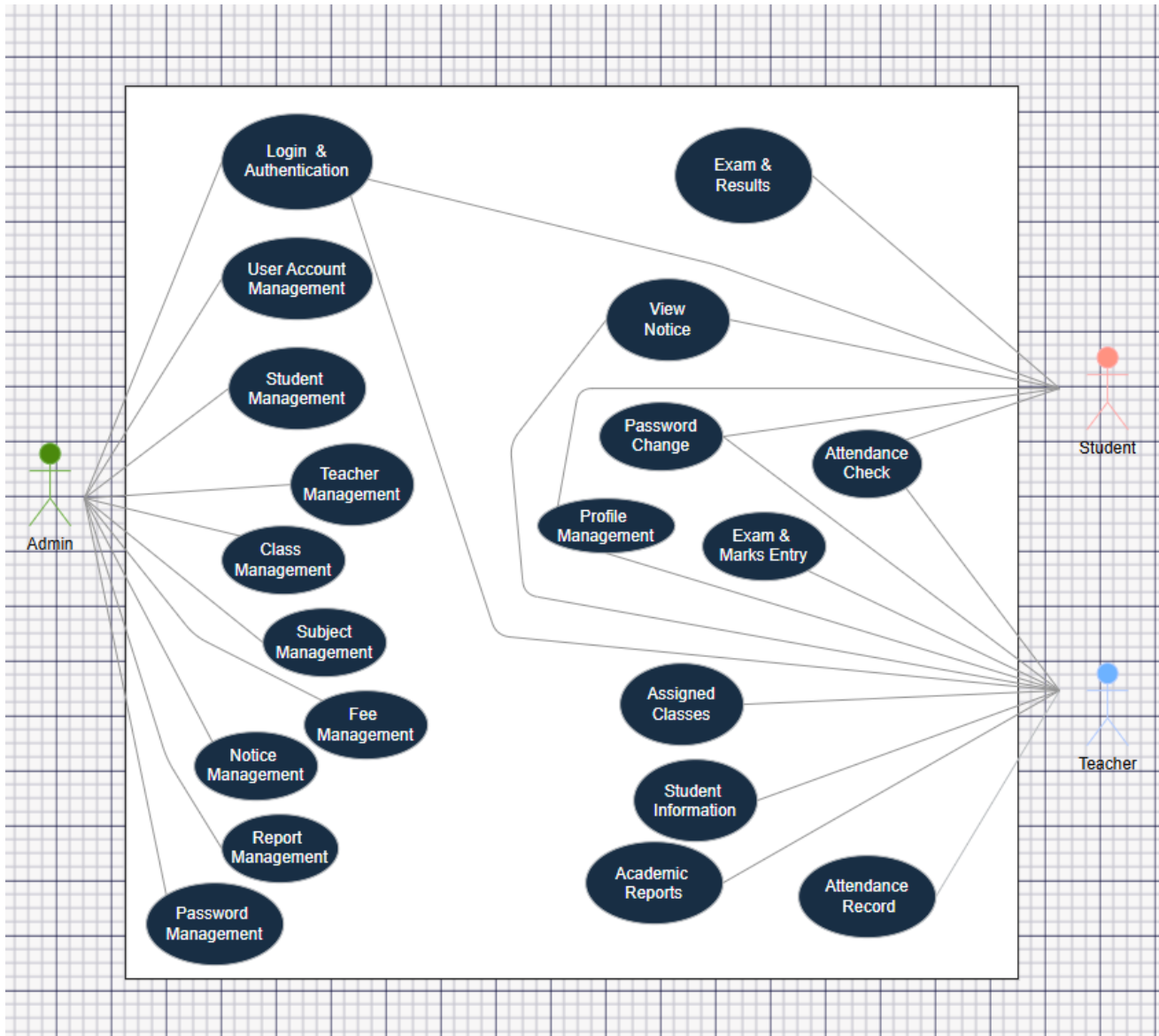


Figure 1: Use Case Diagram of the School Management System

3.1 Use Case Descriptions

ID	Use Case	Actors	Description
UC01	Login / Authentication	Admin, Teacher, Student	Authenticate credentials and provide role-based access.
UC02	User Account Management	Admin	Manage user accounts and control authorized system access.
UC03	Student Management	Admin	Add, update, delete, search, and view student records.
UC04	Teacher Management	Admin	Add, update, delete, search, and view teacher records.
UC05	Class Management	Admin	Create, update, remove, and view class information.
UC06	Subject Management	Admin	Create, update, remove, and view subject information.
UC07	Fee Management	Admin	Record, update, and manage student fee information.
UC08	Notice Management	Admin	Create, update, and publish school notices.
UC09	Report Management	Admin	Generate or view relevant school reports.
UC10	Password Management	Admin	Allow the Admin to change the administrator account password.
UC11	Exam & Results	Student	Allow the Student to view examination marks and results.
UC12	View Notice	Teacher, Student	Allow the Teacher & Student to view notices and announcements published by the school.
UC13	Password Change	Teacher, Student	Allow the Teacher & Student to change the account password.
UC14	Attendance Check	Teacher, Student	Allow the Teacher & Student to view and check attendance records.

UC15	Profile Management	Teacher, Student	Allow the Teacher & Student to view and manage relevant profile and account information.
UC16	Exam & Marks Entry	Teacher	Allow the Teacher to enter and submit examination marks for students.
UC17	Assigned Classes	Teacher	Allow the Teacher to view classes assigned to them.
UC18	Student Information	Teacher	Allow the Teacher to view relevant information about students in assigned classes.
UC19	Academic Reports	Teacher	Allow the Teacher to view relevant academic reports for assigned students or classes.
UC20	Attendance Record	Teacher	Allow the Teacher to record and view attendance for assigned students.

4. USER STORY TABLE

The following user stories are derived from the finalized use case descriptions and represent the requirements of the Admin, Teacher, and Student roles in the School Management System.

ID	User Story	Priority	Acceptance Criteria
US01	As an Admin, Teacher, or Student, I want to log in to the system so that I can access the functions available to my role.	High	Valid credentials provide role-based access; invalid credentials are rejected.
US02	As an Admin, I want to manage user accounts so that I can control authorized system access.	High	Admin can manage user accounts and authorized access.
US03	As an Admin, I want to manage student records so that student information remains accurate and up to date.	High	Admin can add, update, delete, search, and view student records.
US04	As an Admin, I want to manage teacher records so that teacher information remains accurate and up to date.	High	Admin can add, update, delete, search, and view teacher records.
US05	As an Admin, I want to manage classes so that class information remains organized.	High	Admin can create, update, remove, and view class information.
US06	As an Admin, I want to manage subjects so that subject information remains organized.	High	Admin can create, update, remove, and view subject information.
US07	As an Admin, I want to manage student fees so that fee information is properly maintained.	High	Admin can record, update, and manage student fee information.
US08	As an Admin, I want to manage school notices so that important information can be published to users.	Medium	Admin can create, update, and publish school notices.
US09	As an Admin, I want to manage reports so that I can retrieve relevant school information efficiently.	Medium	Admin can generate or view relevant school reports.
US10	As an Admin, I want to change the administrator password so that my account remains secure.	High	Admin can change the password after successful password validation.
US11	As a Student, I want to view examination results so that I can know my academic performance.	High	Student can view available examination marks and results.
US12	As a Teacher or Student, I want to view school notices so that I can receive important announcements.	Medium	Teacher and Student can view notices published by the school.
US13	As a Teacher or Student, I want to change my account password so that my account remains secure.	High	Teacher and Student can change the password after successful validation.
US14	As a Teacher or Student, I want to check attendance records so that I can monitor attendance information.	Medium	Teacher and Student can view and check the available attendance records.
US15	As a Teacher or Student, I want to manage relevant profile and account information so that my information can be accessed when needed.	Medium	Teacher and Student can view and manage relevant profile and account information.

US16	As a Teacher, I want to enter and submit examination marks so that student examination performance can be recorded.	High	Teacher can enter and submit examination marks for students.
US17	As a Teacher, I want to view my assigned classes so that I know which classes I am responsible for.	High	Teacher can view the classes assigned to them.
US18	As a Teacher, I want to view student information so that I can access information about students in my assigned classes.	High	Teacher can view relevant student information for assigned classes.
US19	As a Teacher, I want to view academic reports so that I can monitor information about assigned students or classes.	Medium	Teacher can view relevant academic reports for assigned students or classes.
US20	As a Teacher, I want to record and view attendance so that attendance for assigned students can be maintained accurately.	High	Teacher can record and view attendance for assigned students.

5. REQUIREMENT TRACEABILITY MATRIX

The requirement traceability matrix connects functional requirements with the corresponding user stories, use cases, and planned test cases. This provides a clear path from a requirement to its validation.

5.1 Functional Requirements

Req. ID	Requirement	User Story	Use Case	Test Case
FR01	System shall authenticate Admin, Teacher, and Student users using valid credentials.	US01	UC01	TC01
FR02	System shall provide role-based access according to the authenticated user's role.	US01	UC01	TC02
FR03	System shall allow the Admin to manage user accounts.	US02	UC02	TC03
FR04	System shall allow the Admin to add, update, delete, search, and view student records.	US03	UC03	TC04
FR05	System shall allow the Admin to add, update, delete, search, and view teacher records.	US04	UC04	TC05
FR06	System shall allow the Admin to create, update, remove, and view class information.	US05	UC05	TC06
FR07	System shall allow the Admin to create, update, remove, and view subject information.	US06	UC06	TC07
FR08	System shall allow the Admin to record, update, and manage student fee information.	US07	UC07	TC08

FR09	System shall allow the Admin to create, update, and publish school notices.	US08	UC08	TC09
FR10	System shall allow the Admin to generate or view relevant school reports.	US09	UC09	TC10
FR11	System shall allow the Admin to change the administrator account password.	US10	UC10	TC11
FR12	System shall allow the Student to view examination marks and results.	US11	UC11	TC12
FR13	System shall allow Teachers and Students to view published school notices.	US12	UC12	TC13
FR14	System shall allow Teachers and Students to change their account passwords.	US13	UC13	TC14
FR15	System shall allow Teachers and Students to view and check attendance records.	US14	UC14	TC15
FR16	System shall allow Teachers and Students to view and manage relevant profile and account information.	US15	UC15	TC16
FR17	System shall allow Teachers to enter and submit examination marks for students.	US16	UC16	TC17
FR18	System shall allow Teachers to view their assigned classes.	US17	UC17	TC18
FR19	System shall allow Teachers to view relevant student information for assigned classes.	US18	UC18	TC19
FR20	System shall allow Teachers to view relevant academic reports for assigned students or classes.	US19	UC19	TC20
FR21	System shall allow Teachers to record and view attendance for assigned students.	US20	UC20	TC21

5.2 Non-Functional Requirements

ID	Category	Requirement
NFR01	Performance	The system should respond to normal user actions quickly; the target response time is within approximately 1 second for ordinary operations.
NFR02	Security	Only authenticated users shall access protected functions, and role-based access shall restrict unauthorized operations.
NFR03	Usability	The interface shall be simple, consistent, and user-friendly for school staff and students.
NFR04	Reliability	The system shall save valid records consistently and minimize loss or corruption of stored information.
NFR05	Maintainability	The application shall be structured so that modules and database operations can be maintained without unnecessary changes to unrelated functions.
NFR06	Data Backup	The system/database should support appropriate backup procedures to reduce the risk of data loss.
NFR07	Scalability	The system should be capable of handling increasing numbers of student, teacher, attendance, result, fee, and notice records.
NFR08	Availability	The application should be available for authorized school users during normal operating periods.
NFR09	Technology Constraint	The system shall be developed using C# Windows Forms and Microsoft SQL Server as defined in the project proposal.

5.3 Acceptance Summary

A requirement will be considered satisfied when the corresponding user story acceptance criteria are met and the linked test case produces the expected result. Authentication and role restrictions must be verified before access to protected modules is considered successful.